

Generation Interconnection Request System Impact Study Report

For Queue Project Q24-2456

Tiefort 100 kV
September 02, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2456. The Interconnection Customer proposes a 5.0 MW solar photovoltaic facility interconnecting at the Tiefort 100 kV substation in the Indiana area. The request was received on September 24, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 473 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Monfort - Ne Harriman 100kV L
Contingency: Monfort Substation - Wabash 230kV L
Loading: 111.8% (Rating: 164 MVA)
DFAX: 8.1%
Pre-project loading: 111.5%
2. Facility: Hubbard - Wrangell Substation 100kV
Contingency: Dekalb - Richland 500kV
Loading: 107.2% (Rating: 198 MVA)
DFAX: 7.7%
Pre-project loading: 107.1%
3. Facility: Lonsdale Substation - Hubbard 100kV L
Contingency: Richland - Dekalb 500kV line
Loading: 106.2% (Rating: 221 MVA)
DFAX: 7.7%
Pre-project loading: 106.0%
4. Facility: Lonsdale - Ne Harriman 100kV L
Contingency: Monfort - Wabash 230kV
Loading: 102.6% (Rating: 155 MVA)
DFAX: 8.1%
Pre-project loading: 102.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Monfort - Ne Harriman 100kV	Rebuild line with higher-capacity conductor	\$2,160,000	1% (\$22,000)
Hubbard - Wrangell 100kV	Reconductor line	\$563,000	2% (\$11,000)
Lonsdale Substation - Hubbard 100kV L	Reconductor line	\$867,000	2% (\$17,000)
Lonsdale - Ne Harriman 100kV L	Reconductor line	\$5,170,000	1% (\$51,000)
	Total allocated to project		\$101,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.